



MMA093 Rigid Body Dynamics

Assignment 2

Supplementary Part

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MMA093 Rigid Body Dynamics

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Problem 4:

0.1 CASE 1: $\mu=0$

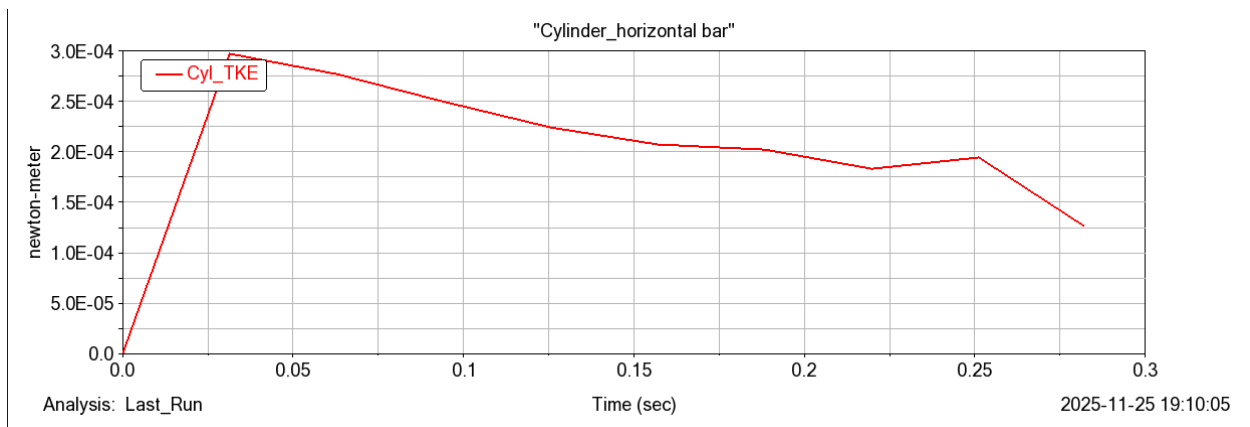


Figure 1: Cylinder Kinetic energy

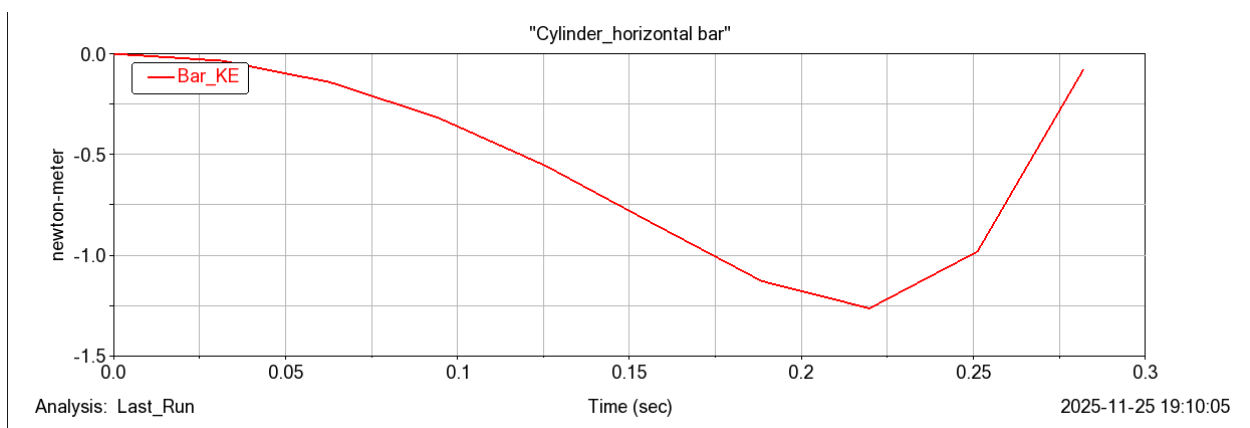


Figure 2: Bar Kinetic Energy

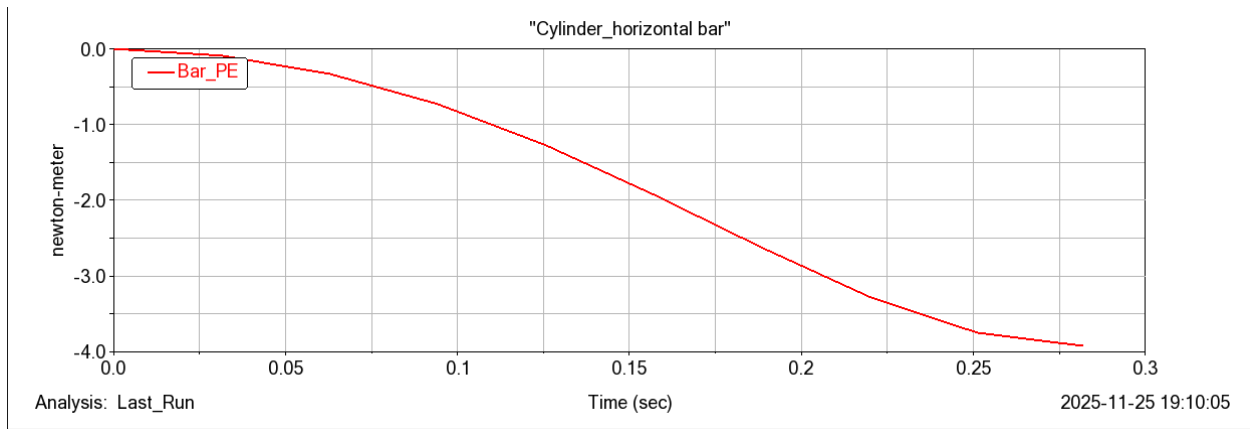


Figure 3: Bar Potential Energy

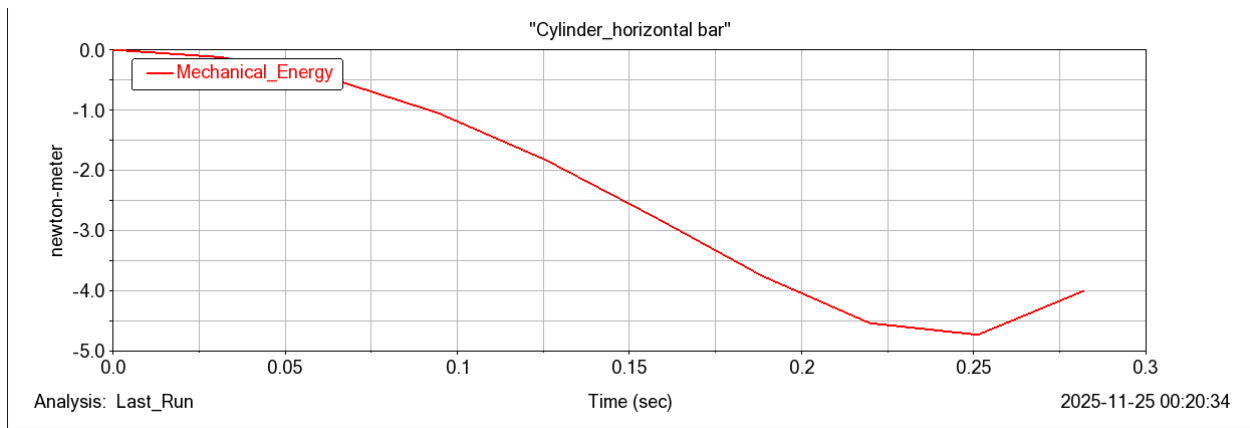


Figure 4: Mechanical Energy

For case 1 the mechanical energy of the system in ADAMS is -3.9982Nm which is close with the analytical -3.9855Nm. With zero friction the contact between cylinder and ground doesn't dissipate any energy. Mechanical energy remains constant throughout the motion.

0.2 CASE 2: $\mu=0.02$

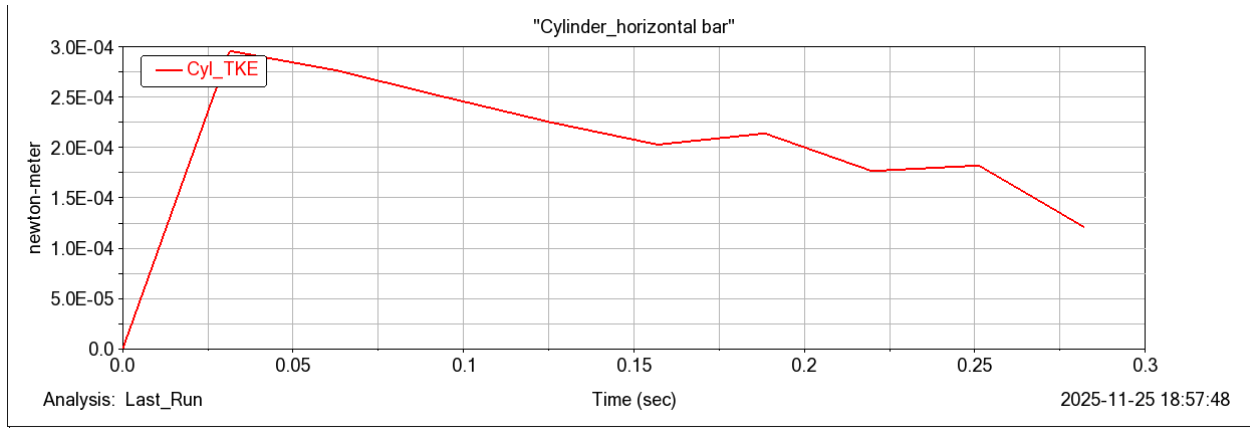


Figure 5: Cylinder Kinetic energy

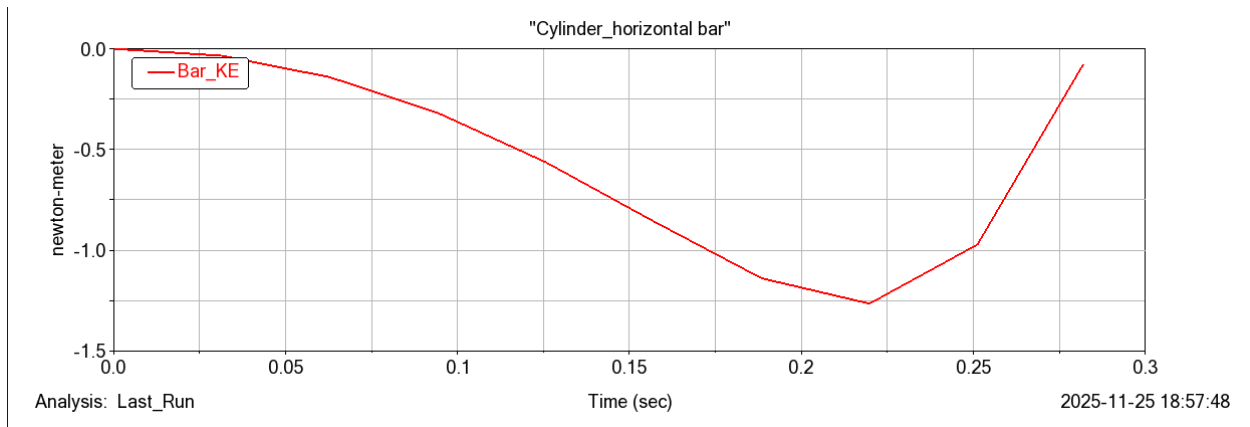


Figure 6: Bar Kinetic Energy

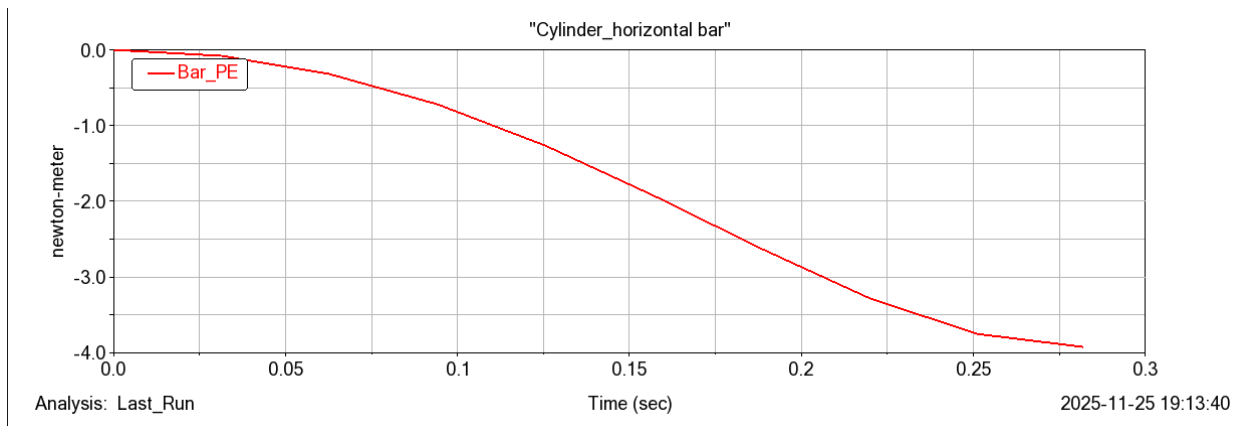


Figure 7: Bar Potential Energy

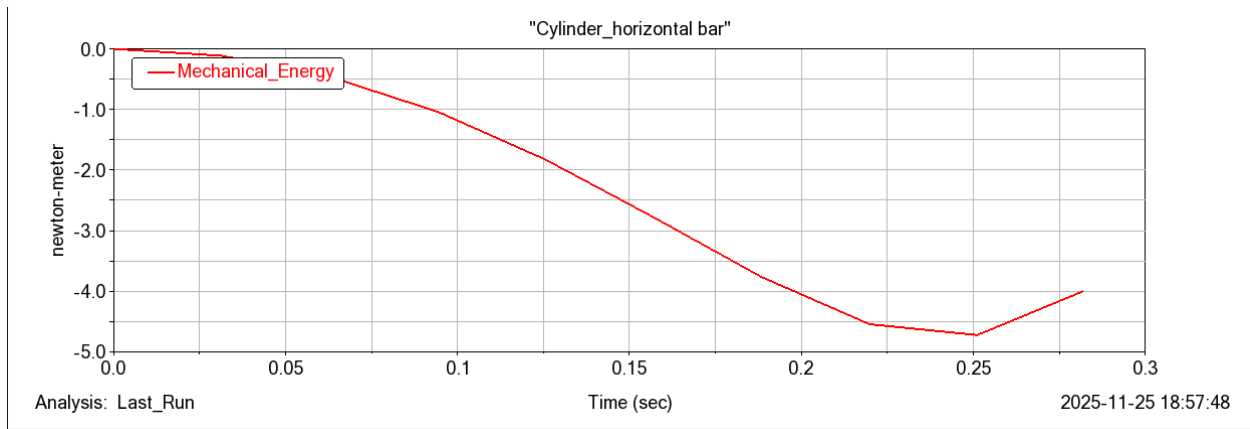


Figure 8: Mechanical Energy

As the $\mu=0.02$ value is small, the cylinder doesn't slip significantly. So the Mechanical energy is same as Case 1 which is -3.9982Nm.

0.3 CASE 3: $\mu = 1.0$

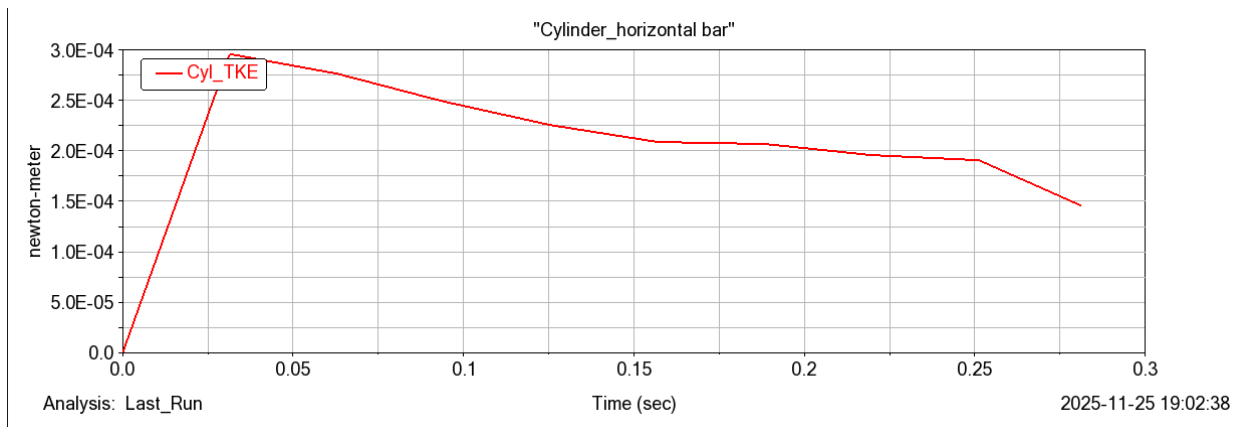


Figure 9: Cylinder Kinetic energy

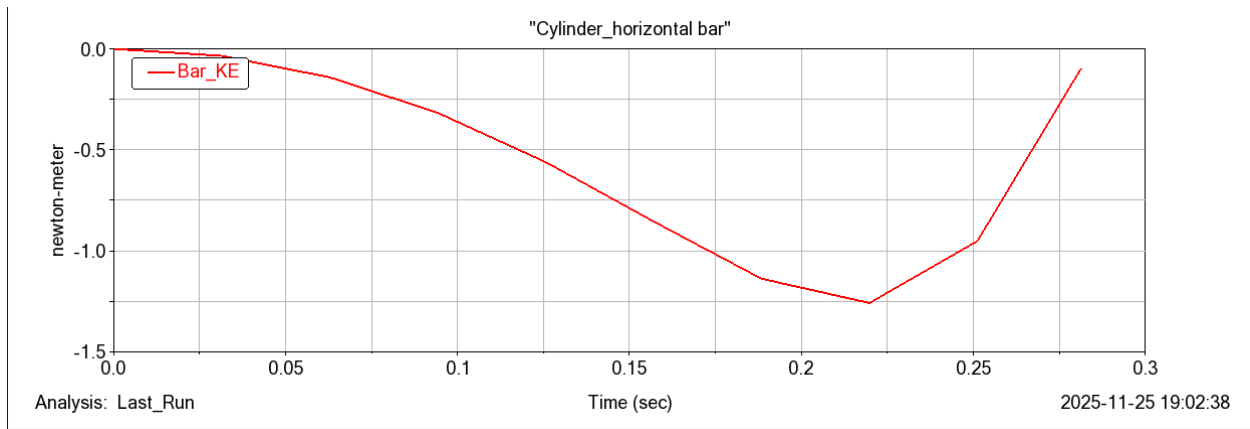


Figure 10: Bar Kinetic Energy

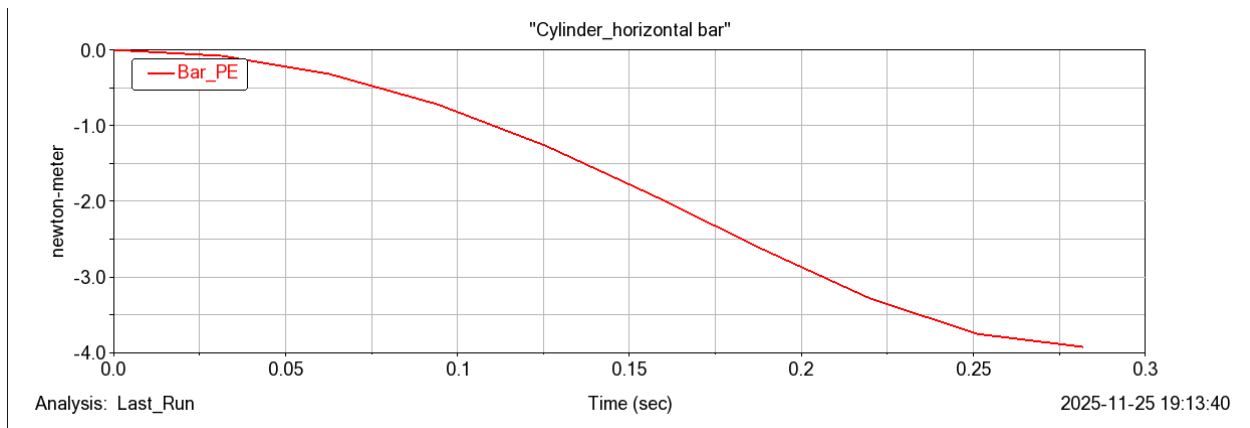


Figure 11: Bar Potential Energy

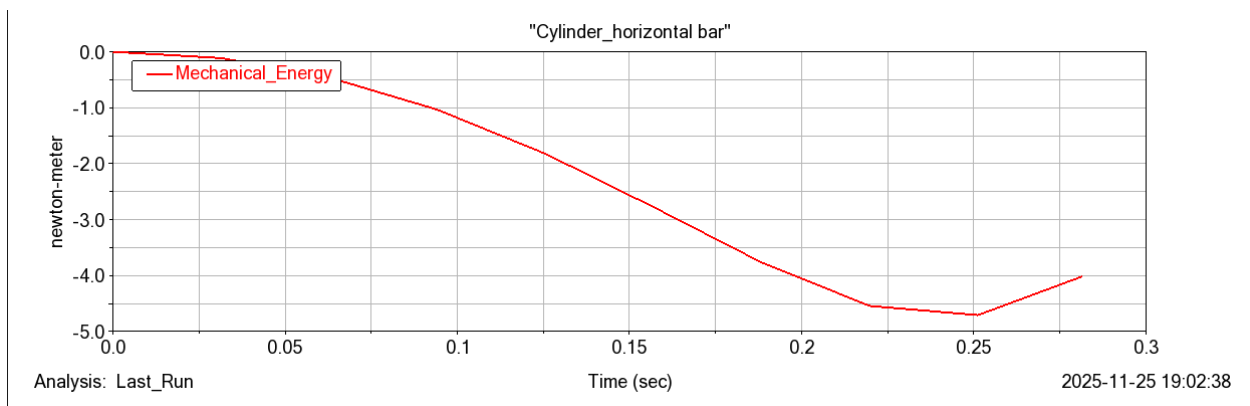


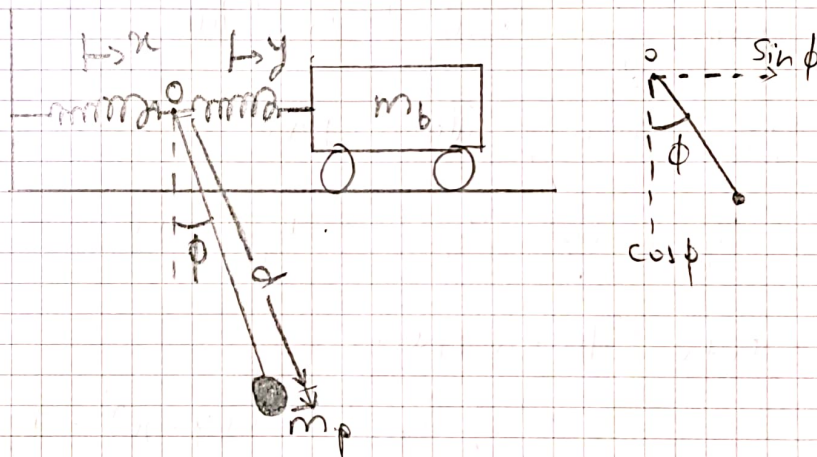
Figure 12: Mechanical Energy

The $\mu = 1$ which is high friction coefficient, the slip occurs between the cylinder and ground due to which mechanical energy decreases to -4.0178Nm.

Problem 7: Problem 5.6 in Bostrom.

Supplementary

Problem 7



Kinetic Energy (T)

Body ~~cart~~ mass m .

Moves along y : $T_{\text{cart}} = T_{\text{body}} = \frac{1}{2} m \dot{y}^2$

Pendulum mass m , length d .

$$x_{\text{pob}} = x + d \sin \phi \quad x_p = x + d \sin \phi$$

$$y_{\text{pob}} = \quad y_p = -d \cos \phi$$

Pendulum velocity

$$\dot{x}_p = \dot{x} + d \cos \phi \dot{\phi}$$

$$\dot{y}_p = d \sin \phi \dot{\phi}$$

$$T_p = \frac{1}{2} m \left[(\dot{x} + d \cos \phi \dot{\phi})^2 + (d \sin \phi \dot{\phi})^2 \right]$$

$$\text{Total Kinetic Energy} = T_{\text{body}} + T_p$$

$$T_{KE} = \frac{1}{2} m \dot{y}^2 + \frac{1}{2} m \left[(\dot{x} + d \cos \phi \dot{\phi})^2 + (d \sin \phi \dot{\phi})^2 \right] \quad \text{--- (1)}$$

Potential Energy.

$$\text{Spring from wall to pendulum pivot} = \frac{1}{2} k x^2$$

$$\text{Spring from pendulum pivot to body} = \frac{1}{2} k (y - x)^2$$

$$V_{\text{pend (gravity)}} = -mgd \cos \phi$$

Total Potential Energy

$$V = \frac{1}{2} k x^2 + \frac{1}{2} k (y - x)^2 - mgd \cos \phi. \quad \text{--- (2)}$$

Lagrangian $L = T - V.$

$$L = \frac{1}{2} m \dot{y}^2 + \frac{1}{2} m \left[(\dot{x} + d \cos \phi \dot{\phi})^2 + (d \sin \phi \dot{\phi})^2 \right] - \frac{1}{2} k x^2 \\ - \frac{1}{2} k (y - x)^2 + mgd \cos \phi.$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right)$$

For x .

$$\frac{\partial L}{\partial x} = -kx + k(y-x).$$

$$\frac{\partial L}{\partial \dot{x}} = m\ddot{x} + md\omega \sin \phi \dot{\phi}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = m\ddot{x} + md(-\sin \phi \dot{\phi}^2 + \omega \sin \phi \ddot{\phi}).$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \left(\frac{\partial L}{\partial x} \right) = 0$$

$$m\ddot{x} + md(\omega \sin \phi \ddot{\phi} - \sin \phi \dot{\phi}^2) + kx - k(y-x) = 0$$

$$m\ddot{x} + md\ddot{\phi} \omega \sin \phi - md\dot{\phi}^2 \sin \phi + kx - k(y-x) = 0$$

For y

$$\frac{\partial L}{\partial y} = k(y-x).$$

$$\frac{\partial L}{\partial \dot{y}} = m\dot{y}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}} \right) - \left(\frac{\partial L}{\partial y} \right) = 0$$

$$= \boxed{m\ddot{y} + k(y-x) = 0}$$

For ϕ .

$$\frac{\partial L}{\partial \phi} = -md\dot{x} \sin \phi \dot{\phi} + mgd \sin \phi.$$

$$\frac{\partial L}{\partial \dot{\phi}} = md\dot{x} \cos \phi + md\ddot{\phi}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \left(\frac{\partial L}{\partial \phi} \right) = 0$$

$$= m d \ddot{\phi} + m d \ddot{x} \cos \phi - \cancel{m d \dot{x} \sin \phi \dot{\phi}} + \cancel{m d \dot{x} \sin \phi \dot{\phi}} - m g d \sin \phi = 0$$

$$= m d \ddot{\phi} + m d \ddot{x} \cos \phi + m g d \sin \phi = 0$$

Divide by d.

$$m d \ddot{\phi} + m d \ddot{x} \cos \phi + m g d \sin \phi = 0$$